

An Undergraduate Internship Project On Data Analysis

By

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Dissertation submitted in partial fulfillment for the degree of Bachelor of Science in Computer Science

Department of Computer Science & Engineering

Independent University, Bangladesh

Attestation

This is, Marufa Akhter Mim, attest to the accuracy of this report, which outlines my assignment at the Aspire to Innovate (A2I) program.

My role involved developing and implementing the SmartDashboard, focusing on data analysis, front-end and UI design, performance optimization, and regulatory compliance. I collaborated with teams and applied data cleaning techniques, contributing to the successful deployment of the platform.

I sincerely thank my supervisor, Md. Ashraful Islam, for their support and guidance. All content in this report is original, with proper citations where applicable.

If you need any additional information or clarification about this project, please do not hesitate to reach out to my supervisor, Md. Ashraful Islam, at A2I. He can be contacted via email at ashraful.islam@a2i.gov.bd.

Marufa Akhter Mim — 15 October, 2024

Signature — Date

Marufa Akhter Mim

Name

Acknowledgement

I would like to extend my heartfelt gratitude to everyone who contributed to making my experience with the SmartDashboard project a rewarding and insightful journey.

First and foremost, I am deeply thankful to the Almighty for granting me the strength, guidance, and opportunity to embark on and successfully complete this project. I also wish to express my sincere appreciation to my family for their unwavering support and encouragement throughout my academic journey. Their constant belief in my abilities has been a source of motivation, inspiring me to overcome challenges and strive for excellence.

I am particularly grateful to the professional i n software development, data analysis, and project management who generously shared their expertise and resources, enriching my learning experience.

A special thanks to Md. Ashraful Islam, my industrial supervisor, for their invaluable support, encouragement, and guidance throughout this project. Their insightful feedback and leadership have pushed me to think critically and strive for excellence. I also extend my thanks to the team members involved in the SmartDashboard project, whose collaboration and dedication have greatly contributed to the project's success.

I am truly grateful for the guidance and support I received during my internship at A2I,and I am confident that this experience will lasting impact on my future aspirations in computer science.

Yours sincerely,

Marufa Akhter Mim

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Letter of Transmittal

Mr. Sarwar Shahidi
Research & Development officer
Department of Computer Science and Engineering
School of Engineering and Computer Science
Independent University, Bangladesh
Subject: Submission of Internship Report for the completion of Graduation

Dear Sir, I am writing to transmit the final report of my undergraduate internship project titled " National Dashboard Framework" as part of the requirement for the degree of Bachelor of Science in Computer Science at Independent University, Bangladesh.

Throughout my internship at Aspire to Innovate (a2i), I had the opportunity to work as a data analyst on the NDF (National Dashboard Framework) project. The report offers a comprehensive overview of the project, including the problem statement, objectives, scope, methodology, and my personal contributions.

I would like to extend my sincere gratitude to my academic supervisor, Mr. Sarwar Shahidi, for his guidance, resources, and support throughout the completion of this internship project. I am also thankful to the Department of Computer Science and Engineering at Independent University, Bangladesh, for providing me with the necessary resources and support to complete this internship project.

I am submitting the final report for your review and evaluation and welcome any feedback or suggestions for improvement.

Sincerely,
Marufa Akhter Mim

Evaluation Committee

Supervision Panel

.....	.....
Academic Supervisor	Industry Supervisor

Panel Members

.....	.....
Panel Member 1	Panel Member 2
.....	.....
Panel Member 3	Panel Member 4

Office Use

.....	.....
Internship Coordinator	Head of the Department

.....
Industry Coordinator of the Department

Abstract

This report encapsulates the culmination of my internship at Aspire to Innovate (a2i), where I served as a data analyst for the National Dashboard Framework (NDF) project. As a key member of the Data Leadership team, I played an integral role in the analysis of project requirements, user interface design (in Figma), and the implementation of enhanced functionalities. My contributions were facilitated by the application of my expertise in SQL & Python, front-end development (using JavaScript, HTML and CSS). Throughout my internship, I was responsible for producing weekly reports and managing a range of internal and external requests, further honing my skills in data analysis and fronted development.

This report provides an in-depth summary of the National Dashboard Framework project, outlining its problem statement, goal, and overall scope. It highlights the use of Agile emphasizes iterative data analysis, continuous feedback loops, and adaptability to evolving project requirements. It promotes collaboration with cross-functional teams, ensuring data insights align with project goals, and facilitates incremental delivery of data-driven solutions to support decision-making throughout the development process. The report also presents a detailed account of my contributions to the project, emphasizing the challenges faced and the approaches used to address them. Overall, my internship at a2i was an immensely fulfilling experience, enabling me to apply my technical expertise to a major real-word initiative while supporting the organization's dedication to innovation, regulatory adherence, and financial integrity.

Keywords— national dashboard framework, data analysis, user interface, front-end development

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Chapter 1

Introduction

1.1 Overview/Background of the Work

In today's digital landscape, the importance of efficient data management and visualization cannot be overstated, particularly for governmental organizations seeking to improve transparency, decision-making, and resource allocation. The "SmartDashboard" project aims to create a centralized data platform for the government, enabling seamless access to diverse datasets and enhancing data-driven governance. This project focuses on developing a user-friendly, secure, and scaleable dashboard that integrates multiple data sources, presenting them in a coherent and actionable format to support various government departments in making informed decisions.

I completed my internship with the a2i Data Team under the Government of Bangladesh, starting on 16 May 2024. My supervisor was Md. Ashraful Islam. During this period, I was actively involved in the development of the "SmartDashboard" project, which is a government-initiated effort to consolidate and visualize data across multiple sectors. My work on the project included requirement analysis, data integration, interface design, and testing to ensure that the dashboard meets the stakeholders' needs and complies with national data security regulations.

Organization Profile a2i (Aspire to Innovate): a2i is a program of the Bangladesh Government aimed at improving public services and governance through digital innovation. It focuses on providing technological solutions to make government services more accessible, efficient, and citizen-centric. The program works across sectors, including education, healthcare, agriculture and governance, using data-driven decision-making tools to foster innovation and improve public service delivery.

Products and Services The SmartDashboard project under the a2i program is designed to serve as a comprehensive data visualization tool. It includes:

- Data Integration: Aggregating data from multiple governmental and non-governmental sources to provide a unified view for decision-makers.

- Data Visualization: Creating interactive dashboards that provide real-time data insights through charts, graphs, and maps.

- Security and Access Control: Implementing strong authentication and authorization methods to safeguard data integrity, security, and compliance with data protection regulations.

- User Management: Overseeing user roles and permissions to regulate access to sensitive data and key features.

Problem Definition The key challenge of the "SmartDashboard" project involves creating an intuitive platform that allows various government departments to visualize and analyze complex datasets effectively. The dashboard must be scalable, secure, and capable of handling large volumes of data while providing a user-friendly experience. The project also needs to address compliance with data protection laws and ensure smooth data integration from diverse sources.

1.2 Objectives

Some of my internship objectives are as follows.

- Real-time data insights: Providing real-time data visualization and insights to support decision-making across various government sectors.

- Interoperability and Consistency: Ensuring consistency and interoperability across all government smart dashboards through the National Dashboard Framework (NDF).

- Monitoring and Evaluation: Systematically planning and monitoring the impact of policies to achieve the Sustainable Development Goals (SDGs) and other national targets.

- Inclusive Development: Facilitating inclusive, data-driven development through tools like the SDG Tracker and My Constituency.

- Crisis Management: Enhancing disaster management and response capabilities, such as improving flood forecasting and warning systems.

1.3 Scopes

In this section, let me breakdown my project scopes.

1. **Data Collection and Cleaning:**

- Source integration: Extract data from various sources and ensure that it is properly integrated into the analysis workflow.

- Data Quality: Implement data cleaning techniques to handle missing values, outliers, and inconsistencies to ensure accurate analysis.

2. **Exploratory Data Analysis (EDA):**

- Descriptive statistics: Use Python libraries (e.g., Pandas, NumPy) to calculate and interpret descriptive statistics.
- Visualization: Create visualizations (e.g., sunburst, bar, pie) to identify patterns, trends, and anomalies in the data.

3. **SQL Query Development:**

- Database Management: Write and optimize SQL queries to retrieve and manipulate data from relational databases.
- Report generation: Generate and validate data reports to support business decisions.

4. **Smart Dashboard Development:**

- Visualization Tools: Design and develop interactive charts and dashboards using the A2I personalized dashboard.
- Data Integration: Ensure that the dashboard effectively integrates data from various sources and displays relevant metrics.

5. **Reporting and documentation:**

- Insight communication: Develop comprehensive reports and presentations to communicate data insights and recommendations.
- Documentation: Maintain clear documentation of data processes, analysis methodologies, and dashboard features.

6. **Collaboration and Feedback:**

- Stakeholder Engagement: Work with stakeholders to understand their data needs and incorporate feedback into analysis and dashboard design.
- Team Coordination: Collaborate with other team members to align data analysis with overall project goals and requirements.

Chapter 2

Literature Review

2.1 Relationship with Undergraduate Studies

The knowledge I gained from my university courses has been incredibly useful in contributing to my internship project. The skills and concepts I learned in my classes are now being applied in practical setting during my internship. Below are some courses that had a direct impact on my internship project:

- CSE 309- Web Application: The expertness I acquired in web application development during this course was crucial for creating the front-end elements of the "Smart¬Dashboard project." The lessons on building user interfaces enabled me to ensure that the front-end design not only adhered to industry standards but also provided an intuitive and user-friendly interface for the bank employees utilizing the application. This foundational knowledge made the development process smoother and more efficient.

- CSE 303- Database Management: The knowledge gained from CSE 303 was instrumental in analyzing data using SQL queries. In this course, I actually learned how to use data maintenance. The course important knowledge in designing efficient database, optimizing queries for data visualization, and implementing strong security and concurrency control strategies. The focus on scalability was particularly valuable, as it helped in designing a dataset architecture processed by the application.

- CSE 213- Object Oriented Programming: This course provides a deep understanding of advanced principles and theories of object-oriented programming. Through learning Java and JavaScript, I have acquired valuable skills that are directly applicable to front-end development, enhancing my ability to create robust and interactive web applications.

- CSE 307- System Analysis and Design: This course enhances my proficiency in user interface design by teaching me how to use "Figma". Mastering "Figma" significantly aids in developing and refining project interfaces, thereby improving the overall design quality and user experience.

- CSE 317- Numerical Methods: This course significantly contributes to meeting the data analysis requirements for this project. Through the lab sessions, I acquired essential skills in data accuracy verification, data visualization, chart creation, data import, data testing, and data cleaning. These competencies align closely with my responsibilities during the internship, making the knowledge gained from this course highly beneficial.

- CSE101- introduction to Computer Programming: The programming knowledge gained in CSE 101 was critical to the successful development of the project. As my introduction to programming, this course provided foundational skills in Python, which made data analysis more manageable during my internship. Without this background, writing code in Python would have been significantly more challenging.

2.2 Related works

Several related efforts in the field of data analysis, user interface design, and web development have contributed to the advancement of tools and techniques that I utilized during my internship. One such example is the "Google Data Studio" platform, which provides a robust environment for data visualization and dashboard creation. Its integration with SQL and other data sources allowed for dynamic and interactive reporting, similar to the 'SmartDashboard' project I worked on.

Another notable effort is the "Tableau" software, known for its powerful data analysis capabilities. Tableau's user-friendly interface and advanced charting options are aligned with the skills I developed in 'Figma' and web development, allowing the creation of intuitive and visually appealing dashboards. The application of data cleaning and visualization techniques in Tableau is directly related to the tasks I performed during my internship, confirming the practical relevance of the knowledge gained from my coursework.

Similarly, BURO handles various types of data, and analyzing these data makes people's lives easier. They actually focused on assisting the poor and rural communities to reduce human poverty and also improve income levels. So they are maintaining different data for different people and using these data they make a chart, those are pretty similar to the project where I worked.

Chapter 3

Project Management & Financing

3.1 Work Breakdown Structure

The Structure for the Work Breakdown (WBS) for the "SmartDashboard" project provides a comprehensive framework for efficiently managing the various tasks required to develop this data-centric application. By not dividing the project into different, manageable parts, the team can tackle different section simultaneously, optimizing workflow and efficiency, thereby boosting efficiency and productivity. The primary goal of the WBS is to streamline project management and ensure a smooth development process. Our WBS is organized into five key phases: requirement analysis, design, development, testing, and deployment.

- Requirement Analysis: The initial phase focuses on gathering and analyzing stakeholder requirements, identifying data sources, and determining the key metrics and functionalities that the "SmartDashboard" must support.

- Design:In this phase, the tasks include developing the system architecture, designing user interfaces using tools like "Figma", and planning the integration of SQL queries and data visualizations.

- Development: The development phase focuses on configuring the development environment, writing Python scripts for data analysis, creating front-end components using JavaScript, HTML, and CSS, and integrating the A2I personalized dashboard for real-time data visualization.

- Testing: During testing phase, is dedicated to validating the dashboard's functionality and integrity. This includes testing the accuracy of data visualizations, the responsiveness of the user interface, and the overall performance of the application.

- Deployment: The final phase ensures a seamless transition to a live environment, including the deployment of the dashboard, monitoring its performance, and making any necessary adjustments based on user feedback.

This structured approach not only accelerates project completion but also fosters effective collaboration within the team, contributing to the overall success of the "SmartDashboard" project.

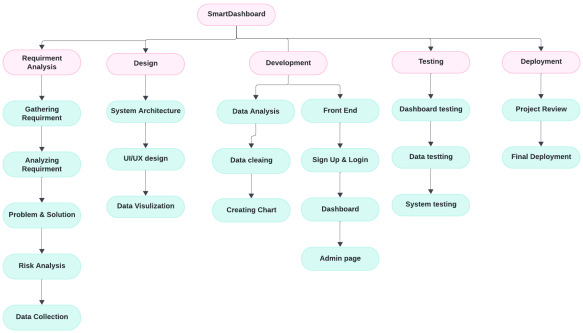


Figure 3.1: Work Breakdown Structure

3.2 Process/Activity wise Time Distribution

The project outlined in the WBS is divided into five key phases. There is a certain amount of time allotted to each step. Breaking down tasks into smaller components and assigning deadlines ensure that the project is completed efficiently. Without time constraints, the project might extend beyond the necessary duration. The project is planned to be completed within 48 working days, divided into five key phase. The Requirement Analysis phase takes 3 days to gather and document user needs. The Design phase requires 5 days to create system architecture and user interface layouts. Development is the most extensive phase, taking 30 days to build the application, including front-end, back-end, and database integration. The testing phase, which lasts 6 days, ensures the application's functionality, performance, and security meet all requirements. Finally, the deployment phase takes 4 days to release the application into the live environment and ensure a smooth transition.

3.3 Gantt Chart

Below is the Gantt chart outlining the timeline for my project, showing the timeline and duration of each phase, from requirement analysis to deployment. The chart visually represents the project schedule, with each task's duration and its starting point in the overall timeline. This helps to identify the sequence of activities and their respective time frames for better project management and tracking.



Figure 3.2: Gantt Chart

3.4 Estimated Costing

The estimated cost of this project covers expenses related to personnel, equipment, and software. Given that this project is being handled by the A2I data team on behalf of the government, the expectations are high, with a strict deadline to complete the project within 2024. The government's primary objective is to consolidate all relevant data into a single, easy-to-understand dashboard that provides accurate and actionable insights. Considering the scope and complexity of the project, the cost estimate is substantial. However, it is important to acknowledge that actual cost may vary due to a variety of factors, including potential changes in project scope, unexpected delays, or unforeseen technical challenges. This variability highlights the importance of diligent budget management and contingency planning to ensure the project stays on schedule and within budget while successfully achieving the desire outcomes.

Chapter 4

Methodology

In the context of this project, a well-structured methodology is essential to ensure the systematic and efficient development of the "SmartDashboard". The chosen method significantly impacts project success, including factors such as timelines, resource management, and the ability to adapt to challenges.

Given the nature of this government project, where the main objective is to create a user-friendly, comprehensive dashboard for data analysis and visualization, the agile methodology has been selected as the preferred approach. Agile is known for its flexibility, iterative development, and emphasis on collaboration, making it particularly suitable for projects with evolving requirements.

Key Aspects of Agile Methodology for This Project:

- **Incremental Delivery:** The project will be developed in small, manageable increments, allowing for continuous assessment and integration of feedback. This approach ensures that government stakeholders can review and provide input at regular intervals, leading to a final product that meets their expectations.

- **Flexibility to Change:** Agile allows the project team to quickly adapt to any changes in requirements or scope, which is crucial given the dynamic nature of data projects. This flexibility minimizes the risk of costly delays and ensures that the project remains aligned with the government's evolving needs.

- **Close Collaboration:** Agile promotes strong collaboration between cross-functional teams, including data analysts, developers, designers, and stakeholders. Regular communication and feedback loops are established to ensure that all aspects of the project are aligned and that any issues are promptly addressed.

- **Continuous Feedback:** By involving stakeholders throughout the project, the team can gather continuous feedback, which is then used to refine and improve the dashboard. This iterative approach enhances the quality of the final product and ensures it meets the desired outcomes.

- **Adaptation to Business Needs:** Agile's iterative nature allows for ongoing adjustments based on the latest data and user feedback, ensuring that the dashboard remains relevant and effective in meeting the government's objectives.

By adopting the Agile methodology, the project team is focused on delivering a high-quality, adaptable dashboard that meets the stringent requirements of the government while ensuring timely completion within the 2024 deadline. This approach not only improves the efficiency of the project, but also ensures that the final product is user-centric, responsive, and capable of providing accurate and actionable data insights.

Chapter 5

Body of the Project

This chapter present detailed documentation that covers various aspects of the project, including the work description, requirement analysis, system analysis, development, testing and design phase. These detailed provided a better understanding of the project's scope and execution.

5.1 Work Description

The SmartDashboard is an advanced data visualization and analysis platform designed to assist government stakeholders in consolidating and interpreting data from a variety of sources. Its primary goal is to simplify the process of data interpretation, providing users with a straightforward interface to access accurate, actionable insights. By integrating multiple data sources into a single, user-friendly interface, SmartDashboard allows users to view, analyze, and understand complex datasets efficiently.

The dashboard is developed with a user-centric approach, with a clean and straightforward interface that enables users to navigate through various data visualizations and reports in a simple and efficient way. Built using modern web technologies, the project uses Python for data analysis, SQL for database management, and a combination of React, HTML, and CSS for front-end development. The system is designed to be scalable and adaptable, capable of managing large volumes of data and delivering real-time insights.

Development life cycle of the SmartDashboard the project team is responsible, which includes requirements gathering and design to, development, testing, and deployment. The team members commitment is focused on ensuring that the final product aligns with the specific needs of government stakeholders, providing a reliable tool for data-driven decision making. The SmartDashboard will not only meet industry standards, but will also offer a responsive and adaptive user experience, ensuring that the application can evolve with future needs and integrate with other systems as required. The development team is dedicated to providing a high-quality solution that meets government requirements and improves data accessibility and interpretation between departments.

5.2 Requirement Analysis

Analysis the requirement phase is difficult for identifying both the functional and non-functional requirement of the SmartDashboard. This stage involves gathering, documenting, and analyzing system needs and constraints to guarantee the solution meets expectations of government stakeholders, providing a reliable and user-friendly platform for data analysis and visualization.

Rich Picture

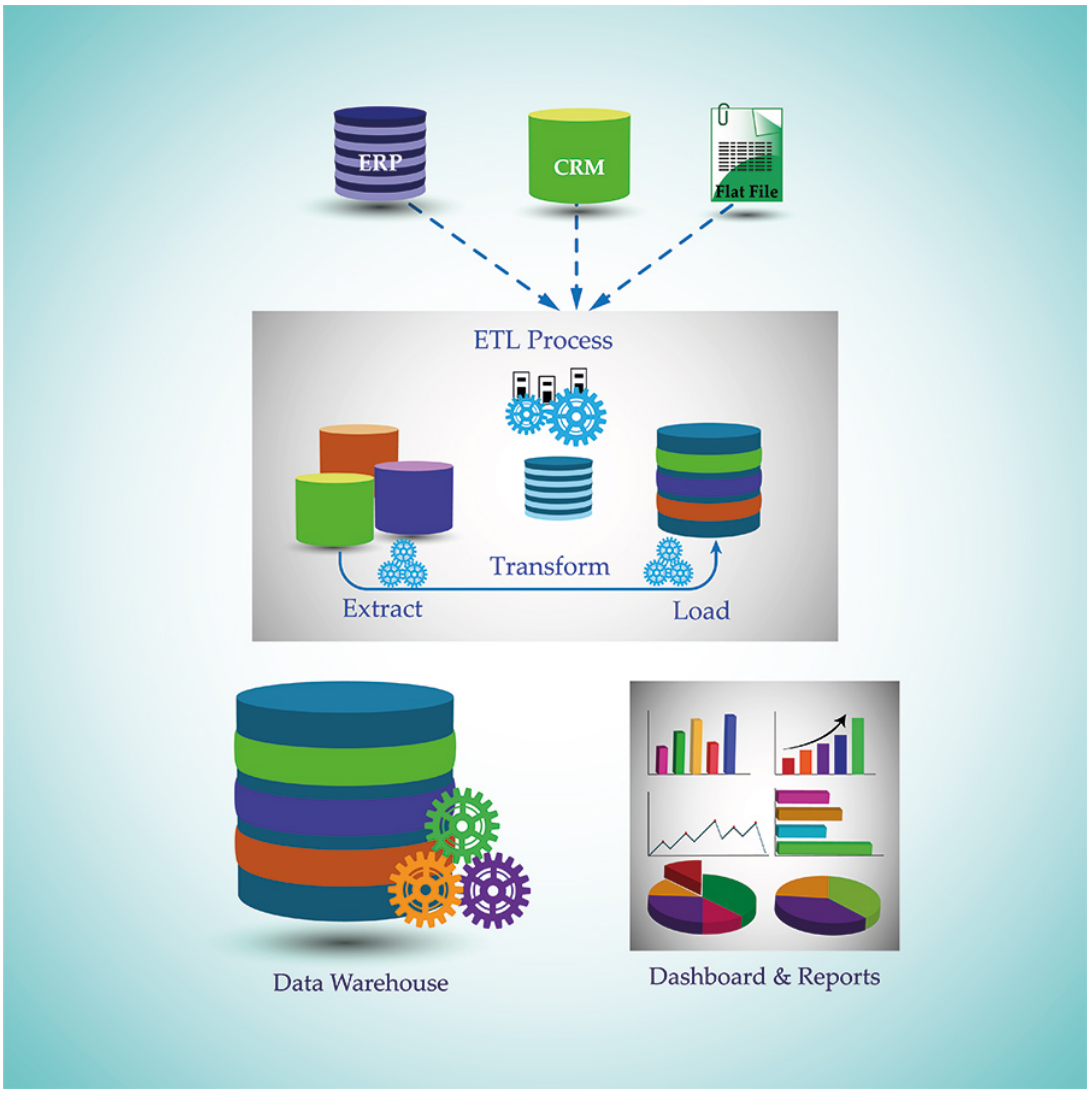


Figure 5.1: Rich Picture for data analysis

Functional and Non-Functional Requirements

Functional Requirements:

- Data Integration: The method must support the integration of data from multiple sources,

including government databases, external APIs, and manual uploads, to provide a comprehensive overview of relevant information.

- Data Visualization: Users should have the ability to create and customize charts, graphs, and reports based on the integrated data, with options for different visualization types (e.g., bar charts, pie charts, line graphs).
- User Roles and Access Control: The platform should support different user roles with appropriate access control; ensuring users only access and modify data relevant to their permissions.
- Search and Filter: The dashboard should allow users to search and filter data easily and quickly locate specific information or trends.
- Exporting Data: Additionally, the system should support exporting visualization and reports in various formats (e.g., PDF, Excel) to facilities sharing and further analysis.
- User Interaction: The interface should offer interactive features, such as drill-down capabilities, to allow users to explore data in more detail.

Non-Functional Requirements:

- Performance:The dashboard must efficiently handle large data volumes and delivering real-time insights without significant delays, ensuring a smooth user experience even with high traffic.
- Security: Robust security protocols are essential to protect sensitive data, including encryption, secure access controls, and compliance with government data protection standards.
- Usability: The user interface must be intuitive and easy to use, enabling users to navigate & operate the dashboard effectively with minimal training.
- Reliability: This system handing effectively error to ensure continuous operation with minimal downtime.
- Scalability:The process must be designed to scale efficiently, accommodating increasing data volumes and user demands without compromising performance.
- Auditability: Comprehensive audit logs must be maintained to track user actions, data modifications, and system changes, ensuring accountability and compliance with regulatory requirements.

The requirement analysis phase ensures that the SmartDashboard is developed with a clear understanding of both functional and nonfunctional requirements, aligning with the government's objectives to deliver a secure, efficient, and scalable data analysis and visualization platform.

5.3 System Analysis

The system analysis stage suggest a detailed summing up the SmartDashboard project to ensure that it aligns with the defined requisite and intention. This stage consists of a six-element analysis, feasibility analysis, problem-solving analysis, and an assessment of effect and constraints.

5.3.1 Six Element Analysis

The six-element examination offers a diversified framework to evaluate the SmartDashboard project, breaking the project down into six key components: inputs, outputs, process, feedback, constraint and environment. Each component plays a vital role in ensuring the successful development and operation of the SmartDashboard, which is designed to provide government stakeholders with a powerful tool for data analysis and visualization.

1. **Inputs:**

- Data Sources:The primary inputs consists of various datasets from government databases, external APIs, and manual data uploads. These datasets may consist of financial records, demographic information, and other critical data required for analysis.

- User Requirements: Key inputs also include the requirements collected from government stakeholders, including the need for specific data visualizations, user roles, and reporting formats.

2. **Processes:**

- Data Integration and Cleaning: Involve integrating data from multiple sources, cleaning and preprocessing it to ensure accuracy, and preparing it for analysis.

- Data Visualization: Converts raw data into meaningful visual formats, such as charts, graphs, and reports that can be easily interpreted by the end-users.

- User Interaction: Allows users to engage with the data, apply filters, drill down into details, and generate customized reports.

3. **Outputs:**

- Visual Reports and Dashboards: The primary outputs are the interactive dashboards and visual reports that display the analyzed data in a user-friendly format.

- Exportable Files: Providing the ability to export data and visualizations in formats such as PDF or Excel for offline analysis and sharing.

4. **Feedback:**

- User Feedback: Feedback mechanisms are integrated into the system to gather user information on the effectiveness and usability of the dashboard. This feedback is crucial for ongoing improvements and updates.
- System alerts: Automated alerts and notifications provide feedback on data irregularities, system performance issues, or necessary actions to ensure timely resolution.

5. **Control:**

- Access Control: Implement user roles and permissions to manage access to sensitive data and specific functionalities within the dashboard.
- Data Validation: Ensure data integrity through validation checks and error handling during data integration and processing.
- Regulatory compliance: Complies with government regulations and standards for data security, privacy, and reporting.

6. **Environment:**

- Hardware: Includes the physical infrastructure needed to run the SmartDashboard, such as servers, storage, and network components.
- Software:Comprises the software stack includes Python for data analysis, SQL for database management, JavaScript, HTML, and CSS for front-end development, and any additional tools or frameworks required to support the dashboard's functionality.
- Operating Environment: Operates within a secure government IT environment, ensuring compliance with security protocols and providing a stable platform for the dashboard's deployment and use.

This six-element analysis provides a structured overview of the SmartDashboard project, emphasizing the critical components necessary for its successful implementation and operation. By carefully considering each element, the project team can ensure that the dashboard meets the government's needs and delivers a robust, scalable solution for data analysis and visualization.

5.3.2 Problem Solution Analysis

Exploration problem solution for the SmartDashboard project involves identifying potential challenges and developing strategies to address them. Key concerns include data accuracy, system performance, and regulatory compliance. Data accuracy is crucial, as inaccurate data could

lead to faulty analysis and decision-making. To solve this, rigorous data validation and cleaning processes are essential to ensure that the data being used is reliable and accurate. System performance is another critical area, especially considering the need to process large datasets quickly and efficiently. Optimizing database queries and improving the server infrastructure are effective solutions to improve processing speed and minimal delays. Finally, regulatory compliance is essential given the sensitive nature of data. The solution involves accomplishing strong security protocols, conducting regular audits, and warranting that all processes are aligned with government regulations and data protection laws. This comprehensive problem-solution analysis ensures that the SmartDashboard project can overcome potential challenges, leading to a reliable, efficient, and compliant data analysis and visualization tool for government stakeholders.

5.3.3 Effect and Constraints Analysis

The effect and constraints analysis of SmartDashboard project highlights its potential to improve government decision-making offering a centralized, easy to use platform for data analysis and visualization. This could lead to more accurate decision, faster processes, and increased stakeholder satisfaction. However, financial, technical, and legal constraints could pose challenges. Managing these limitations is essential to maximize the benefits of the SmartDashboard while ensuring compliance and effectiveness.

5.4 System Design

During system design phase of the SmartDashboard project, a comprehensive plan and architecture are developed to address the requirements identified in earlier stages. This phase includes creating detailed UML models, system architecture designs, and other critical components necessary for the successful development and operation of the application. The design focuses on building robust, scalable, and capable of meeting the project's objectives, including efficient data processing, user-friendly interfaces, and seamless integration with existing infrastructure. The focus is to deliver a well-structured, reliable system that aligns with stakeholder needs and technical constraints.

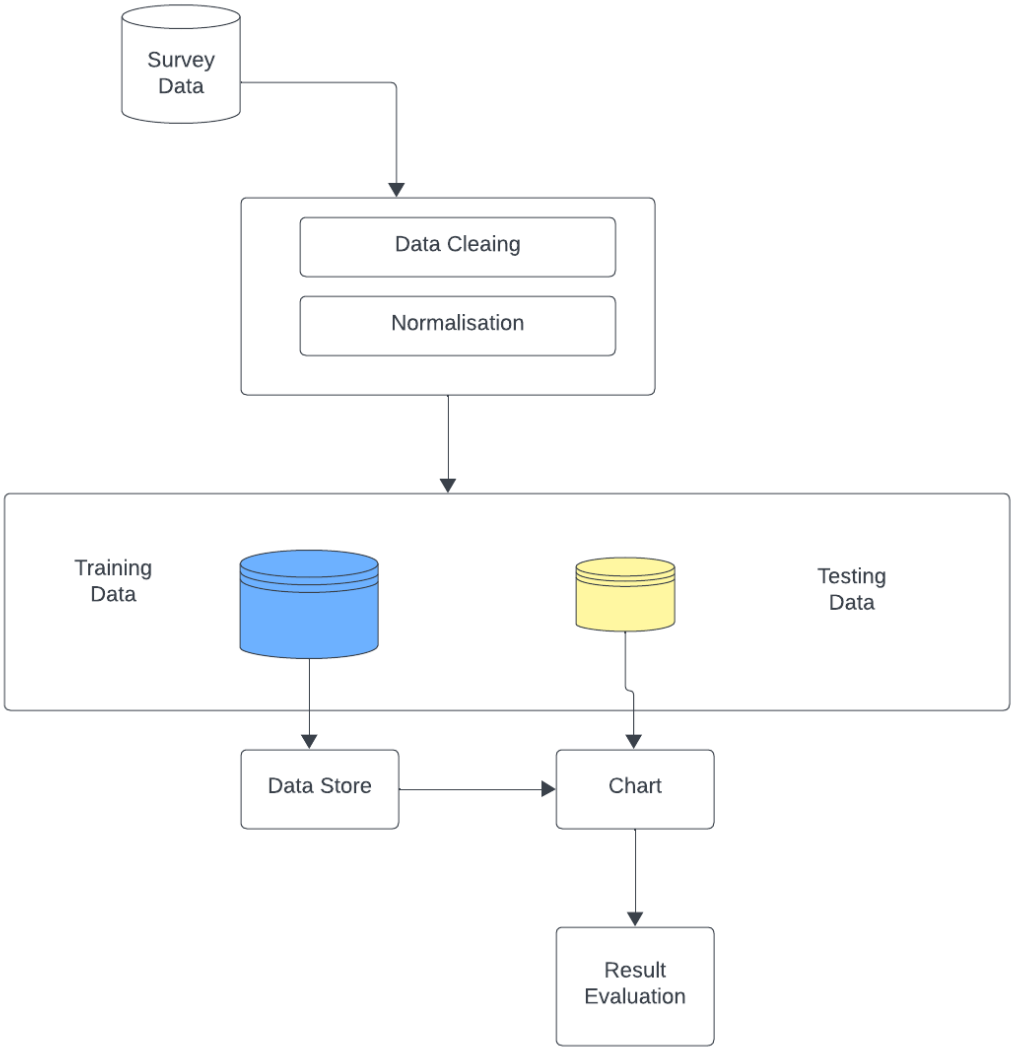
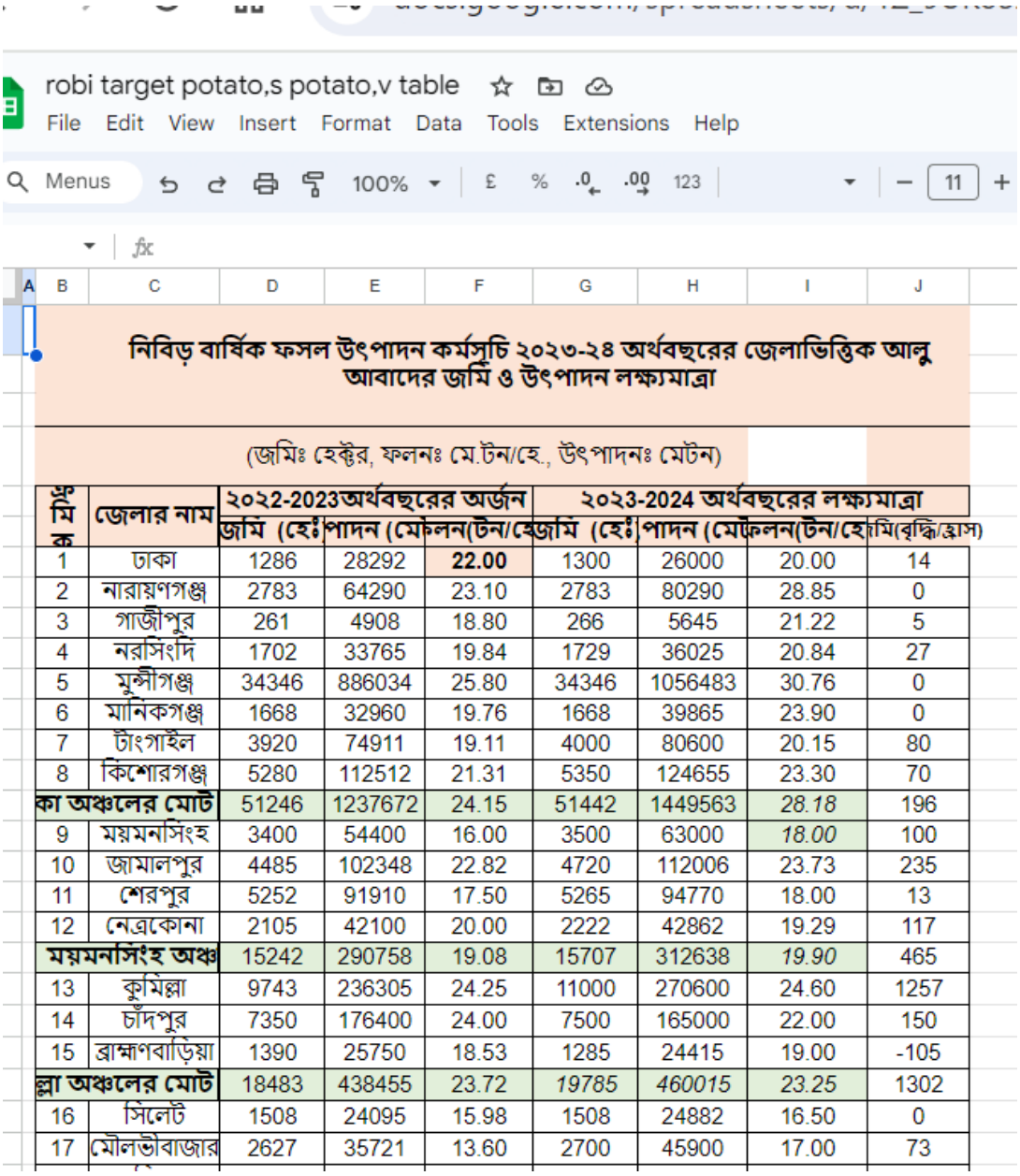


Figure 5.2: System Design

5.5 Development

The development phase involved: UI and Front-End development: using Figma, VsCode.

Input



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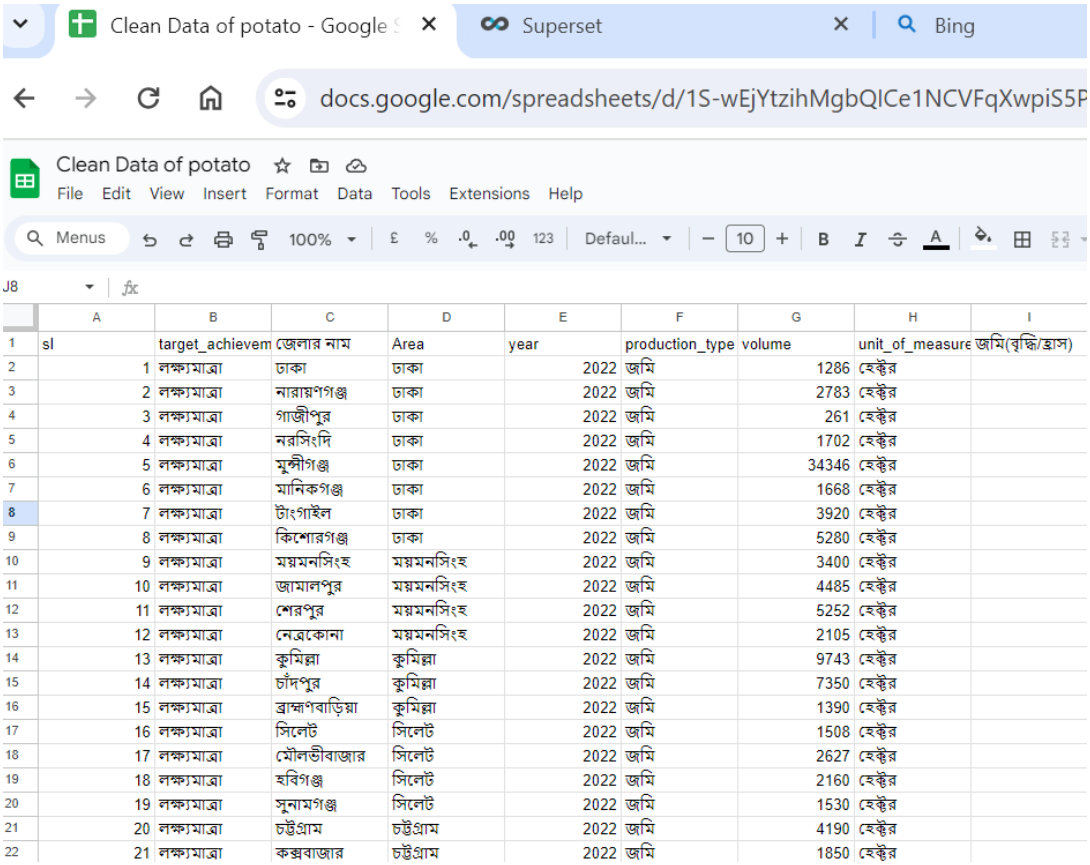
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নিবিড় বার্ষিক ফসল উৎপাদন কর্মসূচি ২০২৩-২৪ অর্থবছরের জেলাভিত্তিক আলু আবাদের জমি ও উৎপাদন লক্ষ্যমাত্রা

(জমিঃ হেক্টর, ফলনঃ মে.টন/হে., উৎপাদনঃ মেটন)

ক্রমিক	জেলার নাম	২০২২-2023অর্থবছরের অর্জন			২০২3-2024 অর্থবছরের লক্ষ্যমাত্রা			
		জমি (হেঃ	পাদন (মেটন	ফলন(টন/হে	জমি (হেঃ	পাদন (মেটন	ফলন(টন/হে	মি(বৃদ্ধি/হ্রাস)
1	ঢাকা	1286	28292	**22.00**	1300	26000	20.00	14
2	নারায়ণগঞ্জ	2783	64290	23.10	2783	80290	28.85	0
3	গাজীপুর	261	4908	18.80	266	5645	21.22	5
4	নরসিংদি	1702	33765	19.84	1729	36025	20.84	27
5	মুন্সীগঞ্জ	34346	886034	25.80	34346	1056483	30.76	0
6	মানিকগঞ্জ	1668	32960	19.76	1668	39865	23.90	0
7	টাংগাইল	3920	74911	19.11	4000	80600	20.15	80
8	কিশোরগঞ্জ	5280	112512	21.31	5350	124655	23.30	70
কা অঞ্চলের মোট		51246	1237672	24.15	51442	1449563	*28.18*	196
9	ময়মনসিংহ	3400	54400	16.00	3500	63000	*18.00*	100
10	জামালপুর	4485	102348	22.82	4720	112006	23.73	235
11	শেরপুর	5252	91910	17.50	5265	94770	18.00	13
12	নেত্রকোনা	2105	42100	20.00	2222	42862	19.29	117
ময়মনসিংহ অঞ্চ		15242	290758	19.08	15707	312638	*19.90*	465
13	কুমিল্লা	9743	236305	24.25	11000	270600	24.60	1257
14	চাঁদপুর	7350	176400	24.00	7500	165000	22.00	150
15	ব্রাহ্মণবাড়িয়া	1390	25750	18.53	1285	24415	19.00	-105
ল্লা অঞ্চলের মোট		18483	438455	23.72	*19785*	*460015*	*23.25*	1302
16	সিলেট	1508	24095	15.98	1508	24882	16.50	0
17	মৌলভীবাজার	2627	35721	13.60	2700	45900	17.00	73

Figure 5.3: Raw Data



Clean Data of potato - Google S | Superset | Bing

docs.google.com/spreadsheets/d/1S-wEjYtzihMgbQICe1NCVFqXwpiS5P

Clean Data of potato

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	A	B	C	D	E	F	G	H	I
1	sl	target_achievem	জেলার নাম	Area	year	production_type	volume	unit_of_measure	জমি(বৃদ্ধি/হ্রাস)
2	1	লক্ষ্যমাত্রা	ঢাকা	ঢাকা	2022	জমি	1286	হেক্টর	
3	2	লক্ষ্যমাত্রা	নারায়ণগঞ্জ	ঢাকা	2022	জমি	2783	হেক্টর	
4	3	লক্ষ্যমাত্রা	গাজীপুর	ঢাকা	2022	জমি	261	হেক্টর	
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6	5	লক্ষ্যমাত্রা	মুন্সীগঞ্জ	ঢাকা	2022	জমি	34346	হেক্টর	
7	6	লক্ষ্যমাত্রা	মানিকগঞ্জ	ঢাকা	2022	জমি	1668	হেক্টর	
8	7	লক্ষ্যমাত্রা	টাংগাইল	ঢাকা	2022	জমি	3920	হেক্টর	
9	8	লক্ষ্যমাত্রা	কিশোরগঞ্জ	ঢাকা	2022	জমি	5280	হেক্টর	
10	9	লক্ষ্যমাত্রা	ময়মনসিংহ	ময়মনসিংহ	2022	জমি	3400	হেক্টর	
11	10	লক্ষ্যমাত্রা	জামালপুর	ময়মনসিংহ	2022	জমি	4485	হেক্টর	
12	11	লক্ষ্যমাত্রা	শেরপুর	ময়মনসিংহ	2022	জমি	5252	হেক্টর	
13	12	লক্ষ্যমাত্রা	নেত্রকোনা	ময়মনসিংহ	2022	জমি	2105	হেক্টর	
14	13	লক্ষ্যমাত্রা	কুমিল্লা	কুমিল্লা	2022	জমি	9743	হেক্টর	
15	14	লক্ষ্যমাত্রা	চাঁদপুর	কুমিল্লা	2022	জমি	7350	হেক্টর	
16	15	লক্ষ্যমাত্রা	ব্রাহ্মণবাড়িয়া	কুমিল্লা	2022	জমি	1390	হেক্টর	
17	16	লক্ষ্যমাত্রা	সিলেট	সিলেট	2022	জমি	1508	হেক্টর	
18	17	লক্ষ্যমাত্রা	মৌলভীবাজার	সিলেট	2022	জমি	2627	হেক্টর	
19	18	লক্ষ্যমাত্রা	হবিগঞ্জ	সিলেট	2022	জমি	2160	হেক্টর	
20	19	লক্ষ্যমাত্রা	সুনামগঞ্জ	সিলেট	2022	জমি	1530	হেক্টর	
21	20	লক্ষ্যমাত্রা	চট্টগ্রাম	চট্টগ্রাম	2022	জমি	4190	হেক্টর	
22	21	লক্ষ্যমাত্রা	কক্সবাজার	চট্টগ্রাম	2022	জমি	1850	হেক্টর	

Figure 5.4: After clean the dataset

```
[ ] import pandas as pd

    filepath_or_buffer = "/content/drive/MyDrive/intern_data/geocode.csv"
    delimiter = '\t'
    header = None

    data = pd.read_csv(filepath_or_buffer, delimiter=delimiter, header=header)

    data_array = data.to_numpy()
```

```
    import numpy as np

    np.random.shuffle(data_array)

    print(data_array)
```

Figure 5.5: Importing Data

```python
import pandas as pd

file_path = '/content/drive/MyDrive/intern_data/geocode.csv'
data = pd.read_csv(file_path)

print("Original Data:")
print(data.head())

cleaned_data = data.dropna()

print("\nCleaned Data:")
print(cleaned_data.head())

cleaned_data.to_csv('/content/drive/MyDrive/intern_data/cleaned_geocode.csv', index=False)
```

Figure 5.6: Cleaning Data

```python
import numpy as np

train_size = int(0.8 * len(data_array))

train_data, test_data = np.split(data_array, [train_size], axis=0)

print("Training Set:\n", train_data)
print("\nTesting Set:\n", test_data)
```

Figure 5.7: Dividing data set into training and testing data

```python
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model = RandomForestClassifier()
model.fit(X_train, y_train)


y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print(f"Model Accuracy: {accuracy * 100:.2f}%")
```

```
Model Accuracy: 76.92%
```

Figure 5.8: Accuracy

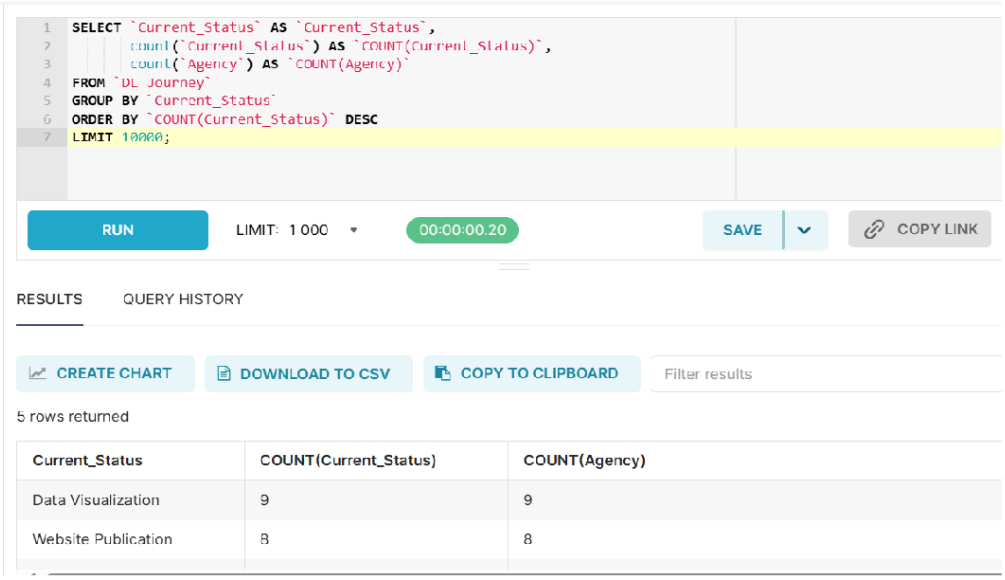


Figure 5.9: SQL query

5.5.1 Output

Data Analysis screenshot from Superset dashboard

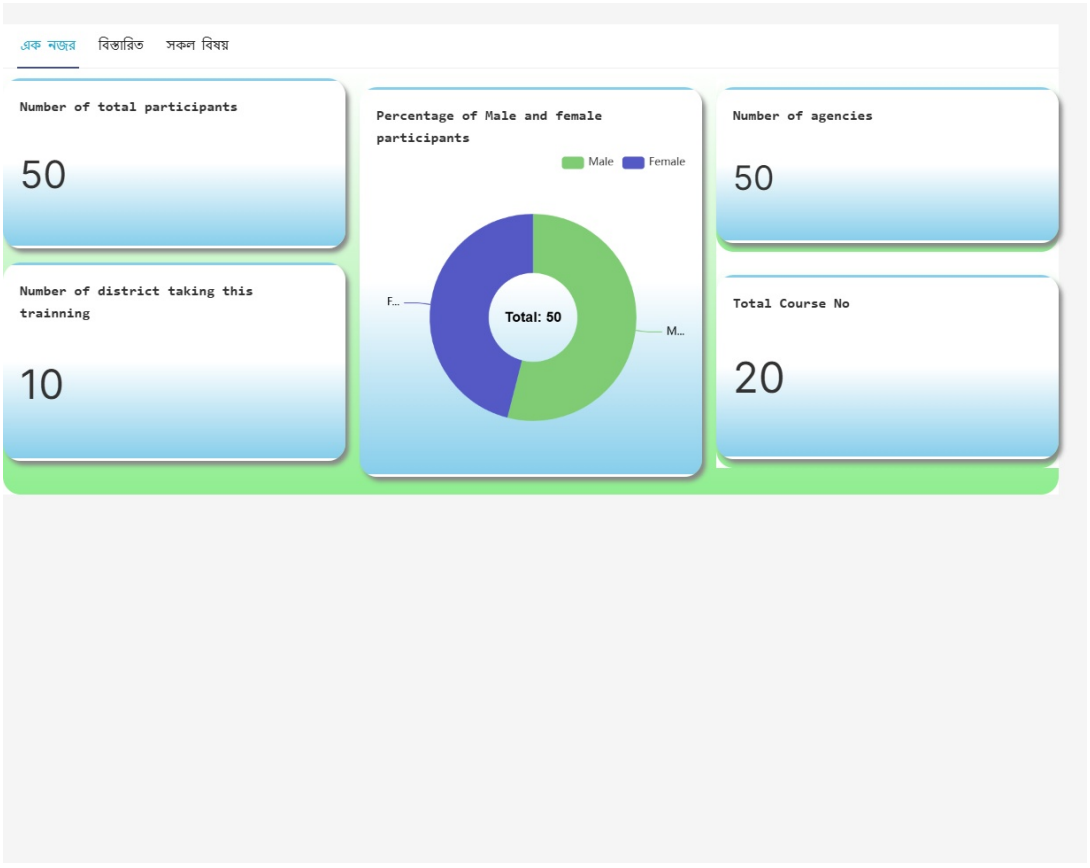


Figure 5.10: Output from dashboard 1

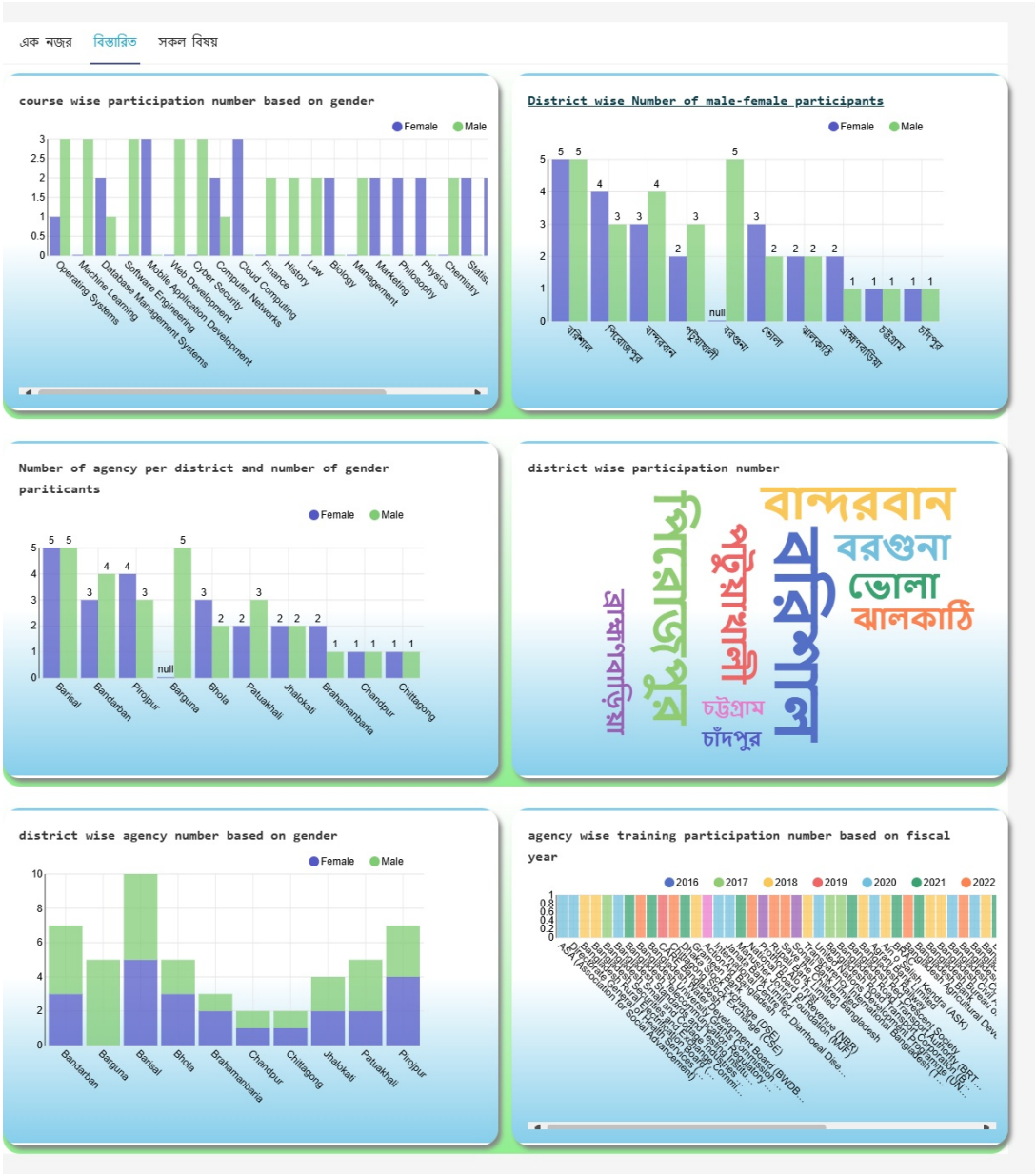


Figure 5.11: Output from dashboard 2

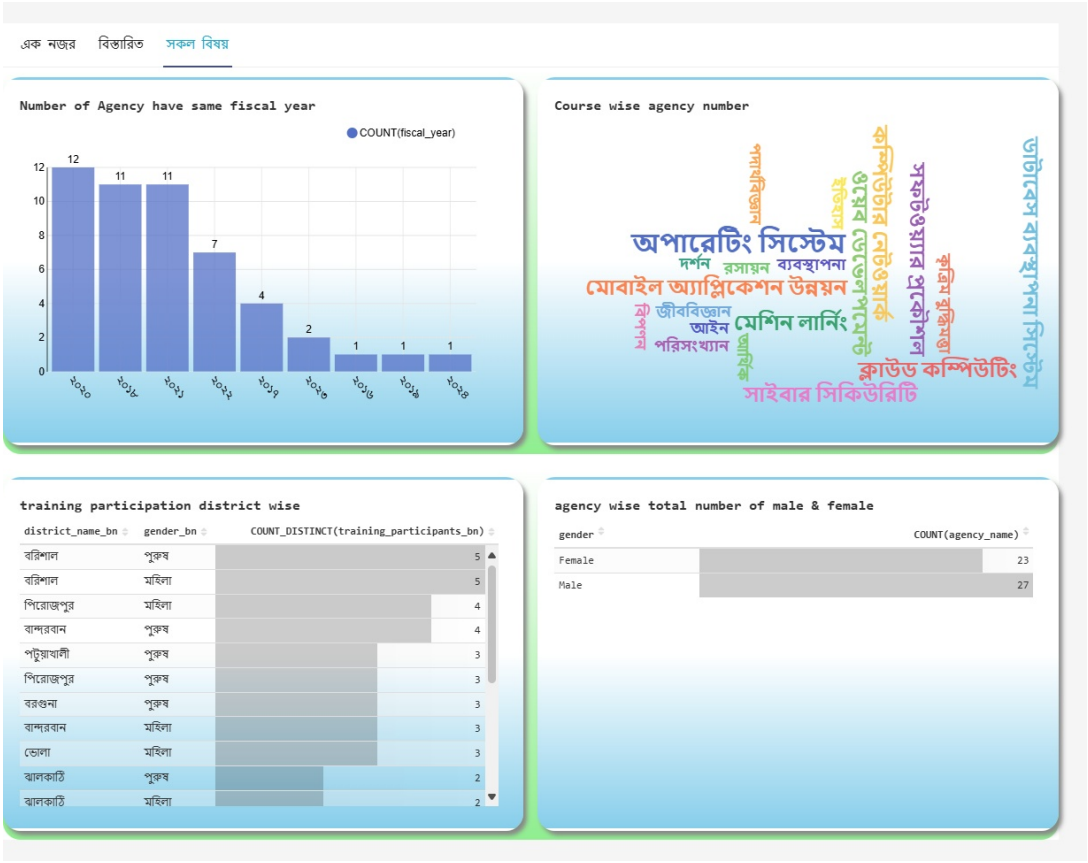


Figure 5.12: Output from dashboard 3



Figure 5.13: Output from dashboard 4

Screen-shoot from Google Colab

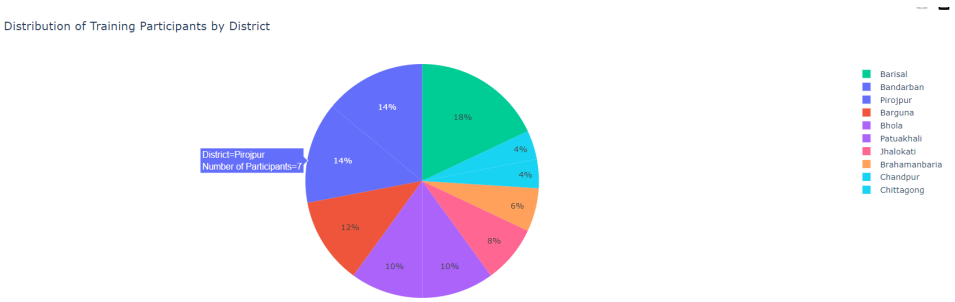


Figure 5.14: Pie chart from colab

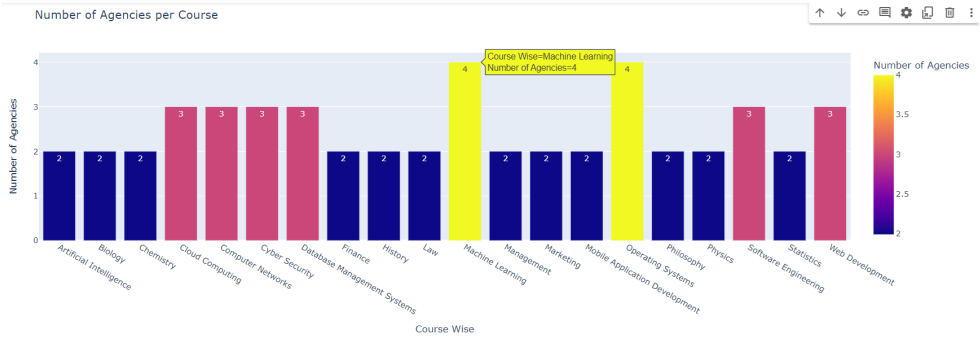


Figure 5.15: Ict bar chart

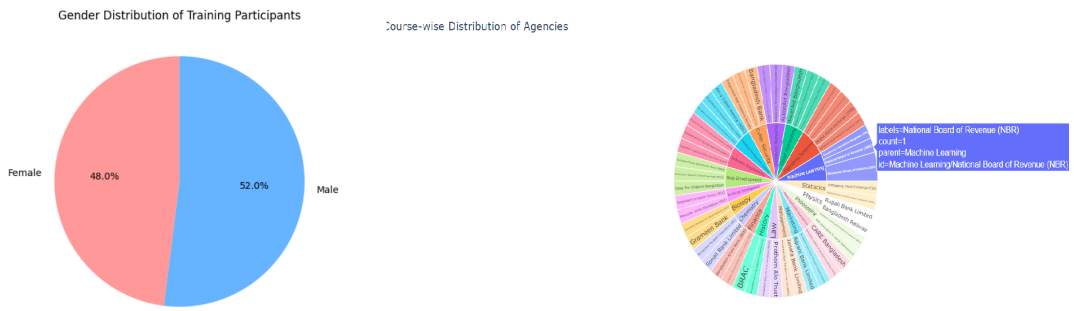


Figure 5.16: Ict pie chart

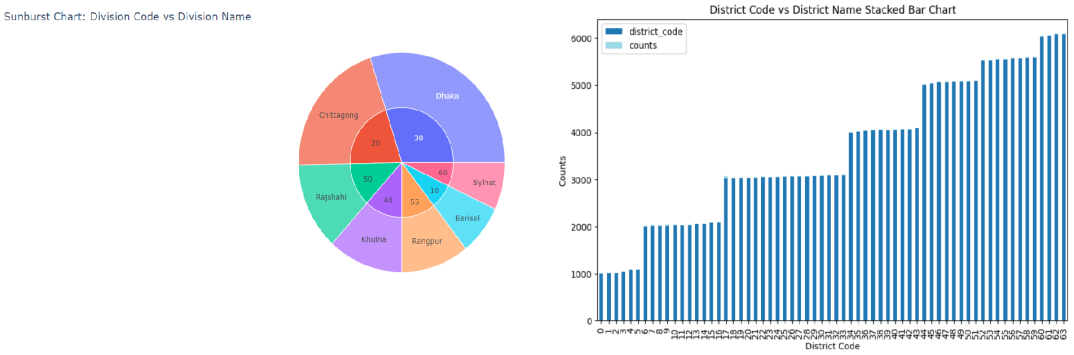


Figure 5.17: Sunburst chart

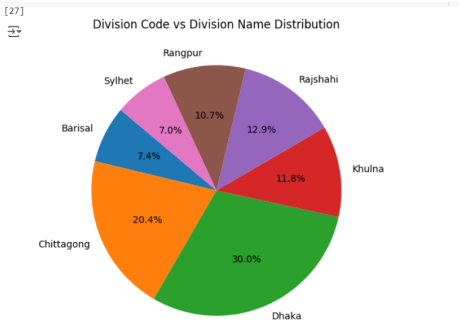


Figure 5.18: Pie chart

Screnshoot from UI design in Figma

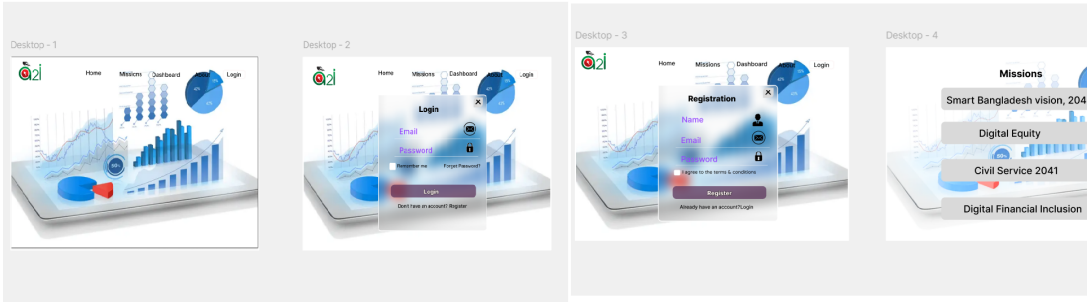


Figure 5.19: Figma 1

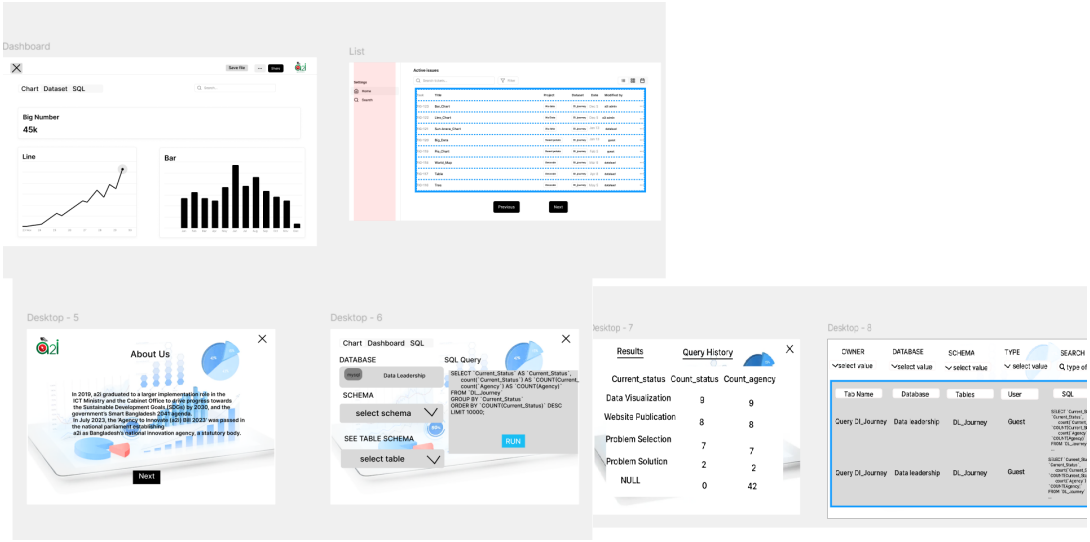


Figure 5.20: Figma 2

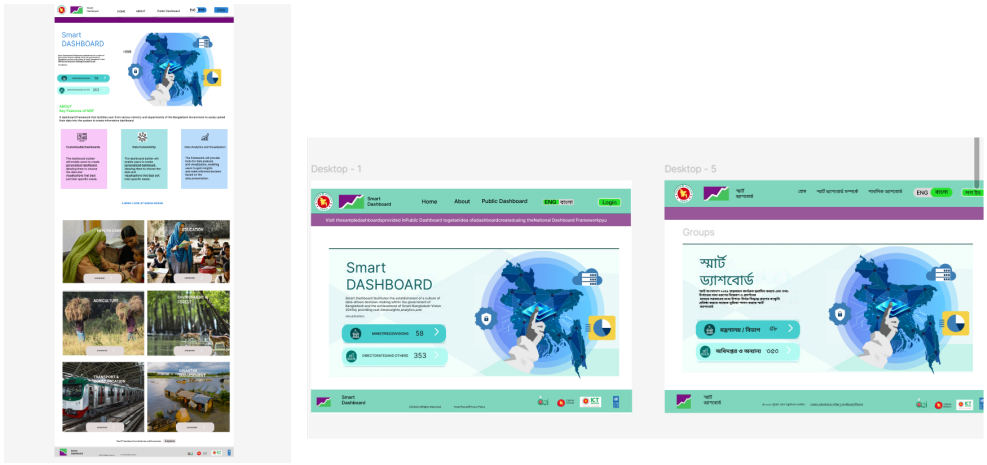


Figure 5.21: Figma 3

Front-end design in the Vs code:

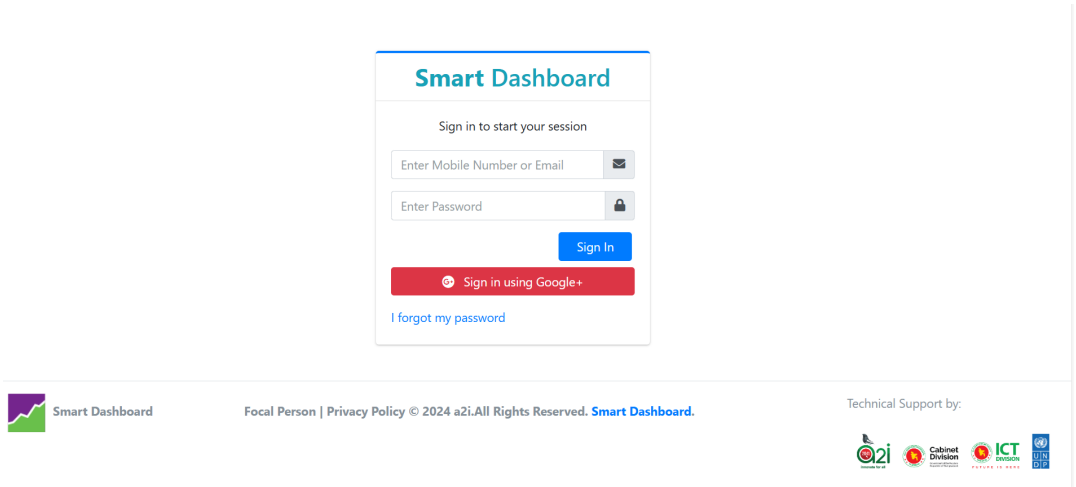


Figure 5.22: Sign in page

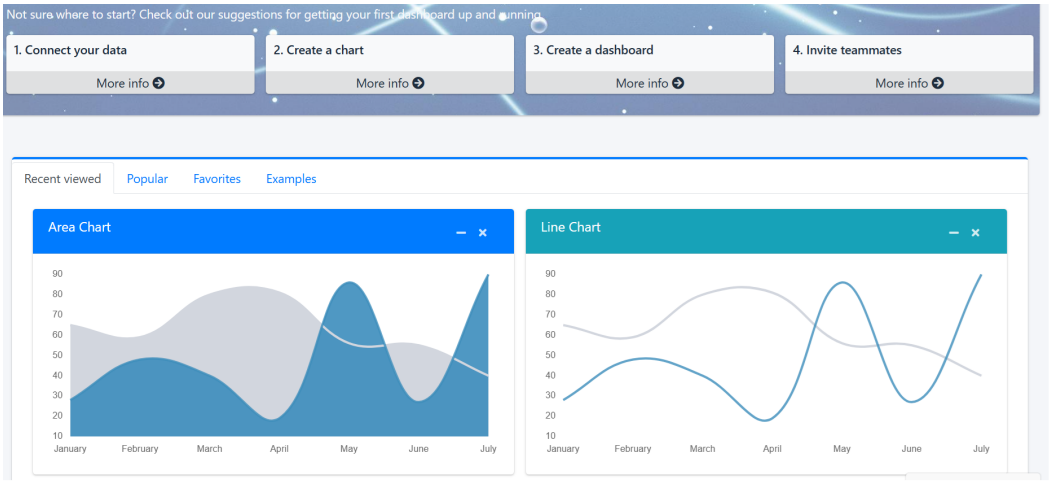


Figure 5.23: Chart page 1

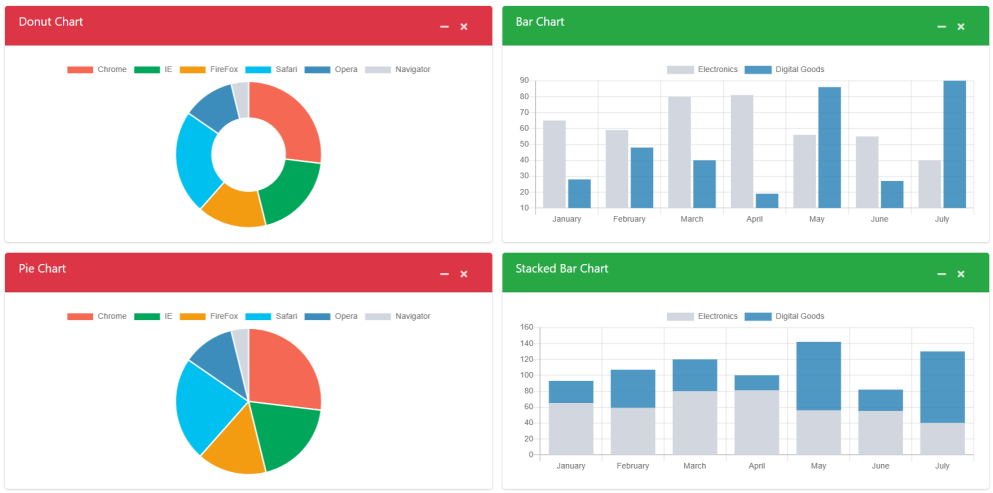


Figure 5.24: Chart page 2

Chapter 6

Results & Analysis

This section evaluated of the result of A2I smart dashboard data analysis front-end development project. It critically examines the practical implication of the project's deployment by analyzing both qualitative and quantitative data.

6.1 Data Integration and Visualization Enhancement

The project successfully integrated various datasets into the SmartDashboard, enhancing data visualization and analytics capabilities for government stakeholders. The unified platform enables real-time access to data and improved decision-making efficiency. These enhancements provide a more comprehensive overview of key information, enabling stakeholders to make faster and better-informed decision.

6.2 Improved User Experience

Significant improvements have been made to the user interface, resulting in a more intuitive and user-friendly experience. The design modifications, informed by user feedback, have simplified the learning process for new users and boosted overall user satisfaction. The responsive design of the dashboard ensures that users can access the system across multiple devices without compromising functionality.

6.3 Optimization of performance and scalability

The project also focused on optimizing performance and ensuring scalability. By optimizing database queries and improving system architecture, the SmartDashboard now efficiently manages larger volumes of data with minimal latency. This scalability ensures that the platform

can handle future increases in data without the need for major redesigns or upgrades.

6.4 Recommendations for Future Work

Moving forward, the following areas are recommended for further attention:

- Security Enhancements:Implementing advanced security protocols to safeguard sensitive data.
- Ongoing System Updates: Continual updates to the software to incorporate new features and maintain optimal performance.
- User Feedback Integration: Regularly gathering and integrating user feedback to guide future enhancements and ensure the platform adapts to evolving needs.

Chapter 7

Project as Engineering Problem Analysis

7.1 Sustainability of the Project/Work

The sustainability of the SmartDashboard project is centered on its capability to remain effective and valuable in the long term within government operations. The platform is designed with scalability and adaptability to adapt to chain data requirements and technological landscapes. Using open-source technologies and a modular system architecture, the project reduces the dependency on proprietary solutions and supports continuous updates and improvements. This approach not only ensures technical longevity, but also facilitates easier maintenance and integration of new features. In addition, the project considers the long-term maintenance and training needs to ensure that users can continue to take advantage of the platform effectively over time.

7.2 Social and Environmental Effects and Analysis

The SmartDashboard project has significant social and environmental implications. Socially, the platform empowers government employees by providing them with a robust tool for data analysis and decision-making, which can lead to better public service outcomes. Enhanced access to accurate data enables more efficient and responsive governance, ultimately benefiting citizens. On the environmental front, the project has been mindful of the energy consumption associated with the infrastructure needed to support the SmartDashboard. By optimizing system performance and minimizing resource consumption, the project aims to reduce its environmental footprint while providing high-quality services.

7.3 Addressing Ethics and Ethical Issues

The SmartDashboard project is deeply committed to addressing ethical considerations, particularly in ensuring the privacy and security of sensitive government data. The project adheres

to stringent data seclusion and protection protocols to ensure that all information remains secure and protected from unauthorized access. Transparency is also a key ethical priority; the project promotes clear and accountable data processing and decision-making practices. In addition, the platform is designed to be accessible to all relevant government departments, ensuring equitable access and use of data analysis tools. By embedding these ethical standards throughout the project's life-cycle, the SmartDashboard aims to uphold integrity, fairness, and social responsibility.

Chapter 8

Lesson Learned

8.1 Problems Faced During this Period

Throughout the evolution of the SmartDashboard project, multiple key challenges arose. One of the most significant difficulties was integrating data sets from various sources, which often came in different formats and contained inconsistent data quality. This discrepancy complicated the data import process and required additional effort to ensure uniformity. In addition, technical issues regarding system performance optimization emerged, particularly when managing large volumes of data. The need to balance system efficiency with high data throughput proved challenging. Compatibility with existing infrastructure and software also caused delays, as adjustments had to be made to avoid disruptions. Furthermore, coordinating across multiple teams while ensuring strict adherence to data protection regulations added complexity to the project timeline.

8.2 Solution of those Problems

To resolve the data integration issues, the development team adopted advanced data cleaning and transformation processes. These methods allowed for the standardization of datasets before being imported into the SmartDashboard, ensuring consistency and accuracy. Performance bottlenecks were addressed by optimizing database queries and utilizing more efficient algorithms, which significantly improved system responsiveness and data processing speed. Compatibility issues were mitigated through extensive testing and system modifications, ensuring seamless integration with the pre-existing infrastructure. To enhance coordination, better communication tools were introduced, which streamline collaboration between teams. Finally, the project maintained full compliance with data protection laws by strictly adhering to regulatory standards and conducting continuous oversight, which minimized the risk of legal complications and ensured ethical handling of sensitive data.

Chapter 9

Future Work & Conclusion

9.1 Future Works

Examining towards future development, there are countless promising directions for the SmartDashboard. A primary focus will be on enhancing the data analytics features by integrating more advanced machine leaning algorithms. These enhancements will enable predictive analytics and trend forecasting, providing users with deeper insights into data patterns. Another important aspect of future work involves expanding the compatibility of the dashboard with a wider range of data sources, allowing a more holistic view of government activities and operations.

Additionally, there is a need to further improve user customization capabilities, and scalability will ensure that the platform can handle increasing data loads as it becomes more widely adopted. Finally, ongoing collection of user feedback and iterative improvements will be crucial to refining the SmartDashboard, ensuring it evolves in line with user needs and technological advancements.

9.2 Conclusion

The SmartDashboard project marks a substantial advance in promoting data-driven decision making within government sectors. By offering a centralized and intuitive platform for analyzing and visualizing complex datasets, the project significantly improves the efficiency and precision of decision-making processes. Despite facing challenges, such as integrating diverse data sources and optimizing system performance, the project team successfully navigated these obstacles with innovative approaches and effective teamwork. As the SmartDashboard continues to develop, it is well-positioned to become an essential tool for government agencies, enabling more informed and effective decision making. The lessons learned and the foundation laid by this project will guide future enhancements, ensuring that the platform remains robust, adaptable, and valuable for addressing the evolving needs of its user in the coming years.

Bibliography

[1] Ben Williamson. Digitizing education governance: Pearson, real-time data analytics, visualization and machine intelligence. *Education Governance and Social Theory: Interdisciplinary Approaches to Research*, pages 21–42, 2018.

[2] Chengqing Zong, Rui Xia, and Jiajun Zhang. *Text data mining*, volume 711. Springer, 2021.

[2],[1]

An Undergraduate Internship/Project on Data Analysis

By

Marufa Akhter Mim

Student ID: **2022320**

Summer, 2024

The student modified the internship final report as per the recommendation made by his or her academic supervisor and/or panel members during final viva, and the department can use this version for achieving.

Signature of the Supervisor

Sarwar Shaihidi
Research & Development officer
Department of Computer Science & Engineering
School of Engineering, Technology & Sciences
Independent University, Bangladesh

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